Development of a Mental Health Chatbot Using Large Language Models for Indonesian Undergraduates

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***Abstract*—Mental health issues among university students are an escalating global problem, especially in Indonesia. Young adults, especially college students, are a vulnerable demographic due to academic pressures, social transitions, and the challenges of independent living. This paper describes the development of a mental health chatbot specifically for Indonesian university students, utilizing large language models (LLMs). The development encompassed three essential phases: establishing a knowledge base, refining the LLM with culturally pertinent Indonesian data, and assessing the system. The findings demonstrated notable enhancements in the chatbot's precision, with training accuracy rising from 0.6 to 0.85 and validation accuracy from 0.5 to 0.79 due to fine-tuning and regularization. User acceptability testing including 58 students resulted in an acceptance percentage of 87.38%, indicating substantial satisfaction with the chatbot's response and usefulness. This research highlights the efficacy of AI-driven interventions in addressing the mental health requirements of Indonesian university students.**

Keywords— Mental Health, Chatbot, Large-Language Model, Student, Psychologist

I. INTRODUCTION

Mental health challenges among university students are a growing concern globally, and Indonesia is no exception. A study of 227 student respondents revealed that approximately 26.9% of them suffered from mild depression, 18.5% from moderate depression, and 9.3% from severe or extreme depression. University students are at a critical juncture in their lives, facing academic pressures, social challenges, and the uncertainties of future careers. These factors can exacerbate pre-existing mental health conditions or lead to the development of new issues such as anxiety, depression, and stress disorders [1].

Latest data from the Indonesia National Adolescent Mental Health Survey (I-NAMHS) reveals that 34.9% of adolescents experience mental health problems, with anxiety and depression being the most common disorders. As these young individuals transition to university life, their mental health challenges often persist, or even worsen, due to increased responsibilities, social expectations, and academic pressures [2].

As awareness of mental health has grown, multiple strategies have been put together to help students deal with stress, anxiety, and other mental health problems. However, limited mental health services, social stigma, and lack of time to access professional help remain significant barriers. In response to these challenges, technology, specifically AI-based chatbots, emerges as a potential solution to provide more accessible and private support [3].

It possesses exceptional capabilities to naturally understand and respond to human language. This technology has proven effective in facilitating interactions that mimic human conversations, making it highly suitable for mental health applications. By utilizing LLMs fine-tuned to customized datasets, the chatbot can adapt to users' language and culture, providing more contextually relevant support. This technology can help students recognize and manage their mental health conditions without facing the stigma or time barriers often associated with traditional mental health services [4].

To ensure meaningful and productive interactions, it is essential to develop a chatbot that adapts to the language and culture of Indonesia. Therefore, this project endeavors to construct a mental health chatbot that not only harnesses artificial intelligence (AI) but also prioritizes a culturally sensitive approach to cater to the unique needs of Indonesian students [5]. Through this chatbot, students can access fast, easy, and anonymous mental health support at any time. A large language model (LLM), fine-tuned on a dataset relevant to Indonesia's social and linguistic context, trains the chatbot. With this approach, the chatbot can provide appropriate advice and even refer users to professional services when needed [6].

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We expect this approach to tackle the primary issues in mental health services in Indonesia, specifically accessibility and stigma. We hope that an LLM-based chatbot will better support students in managing their mental health conditions and pave the way for artificial intelligence-based solutions in Indonesia's mental health field [7].

We cite a recent work called "Chat Counselor: A Large Language Model for Mental Health Support" [8] to bolster our approach. This study focuses on improving an LLM through the use of professional counseling data. Although their system is quite good at extracting important data from actual counseling sessions, our approach has a number of unique benefits. Specifically, we tailor our fine-tuning technique to the language and cultural context of Indonesian students, using a particular customized dataset. This strategy addresses cultural nuances that general fine-tuning methods, such as those used in chat counselors, may overlook, ensuring not only accurate technical performance but also increased relevance and resonance in real-world applications. We provide a solution that is more sensitive to the demands of individual users by incorporating local knowledge into the model, which eventually increases user engagement..

II. RELATED WORK

A. Recent Work of Chatbots in Mental Health

In recent years, AI-enabled chatbots have garnered considerable attention in the mental health field due to their capability to offer scalable, accessible, and confidential support. The shortage of professionals exacerbates the increasing global demand for mental health services, which these systems address.

TABLE I. RELATED WORKS

Aspect	*Comparative Study*			
	[9]	*[8]*	*[10]*	*This Study*
Methodology	Scoping review of mental health chatbot applications	Meta-analysis of CA interventions using sentiment analysis and NLP	Systematic review of clinical trials and empirical studies on AI chatbots	Fine-tuned LLM on local dataset for Indonesian students
Symptoms Addressed	Depression, anxiety, substance use disorders	Depression, anxiety, emotional well-being	Depression, anxiety, adherence to treatment	Stress, anxiety, depression
Application Type	Smartphone and web-based chatbots, mainly text-based	Platforms include smartphones, robots, web, and virtual reality	Therapy, counseling, and self-help interventions	Web-based interface using Streamlit
Advanced Features	AI personalization, mood tracking, regular check-ins	Sentiment analysis, emotion AI, machine learning enhancements	AI-enabled "virtual psychiatrist" models	Cultural adaptation, multi-turn conversation management
User Engagement	Engagement measured by length, duration, module use conversation	Focus on user interactions and mental health moderators like baseline	User satisfaction and therapeutic alliance evaluation	58 student testers with stratified random sampling

Studies suggest that artificial intelligence-based chatbots effectively manage symptoms of depression, anxiety, and other mental health conditions by leveraging advanced technologies such as sentiment analysis, personalized interventions, and real-time mood tracking. [9], [10]. Over the past decade, chatbots have become pivotal in assisting users with mental health challenges. Particularly in regions where access to traditional mental health services is limited, they offer accessible, anonymous, and efficient support. The integration of large language models (LLMs) has further enhanced the potential of chatbots to deliver more personalized and adaptive mental health care [11]. This paper compares these advancements with our system, which leverages fine-tuned LLM and cultural adaptation techniques to provide mental health support for Indonesian university students. The comparison highlights key differences in methodology, symptoms addressed, and user engagement features. Table I has already presented the details.

B. The Importance of Mental Health Support for University Students

Poor mental health can negatively impact students' academic performance, social relationships, and overall well-being. Many struggle in silence without the proper support, which can worsen their conditions over time. Therefore, easily accessible and anonymous mental health support is essential, especially for students who are vulnerable to academic and social pressures. Solutions such as artificial intelligence (AI)-based chatbots can provide fast and responsive support, helping mitigate the stigma associated with seeking help in person.

The use of LLM-based chatbot technology, as discussed in the previous subsection, offers an effective solution in this context. Chatbots trained to understand the context and emotions of users can provide emotionally relevant responses and assist students in managing their conditions [12].

C. LLM Technology in the Context of Mental Health

The development of LLMs, such as GPT-4, has brought significant advancements in mental health chatbots by enhancing their ability to engage with users and understand conversational contexts. LLMs enable chatbots to process natural language effectively, generating empathetic and relevant responses that are crucial in supporting users facing mental health challenges. These models, trained on vast datasets, can recognize informal language, slang, and cultural nuances—elements critical for providing contextually appropriate support in mental health interventions. In situations where immediate and accurate responses are essential, LLMs can detect emotional cues from user inputs, such as anxiety, distress, or depression, and respond accordingly, offering timely assistance [12].

In addition to their linguistic capabilities, LLM-based chatbots offer substantial improvements over traditional rule-based systems by handling more complex and nuanced conversations. By fine-tuning LLMs with culturally and linguistically specific datasets, such as those used in this project, chatbots can provide more personalized responses that reflect the user's emotional state and cultural context [3]. However, despite these advantages, LLM-based chatbots still have limitations in fully replicating the empathy and nuanced understanding that human professionals provide, especially in cases involving severe mental health conditions. Nevertheless, their ability to deliver scalable, anonymous, and accessible mental health support represents a major advancement in regions with limited access to professional mental health services [10].

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III. METHODS

We followed a structured methodology in the development of this mental health chatbot to ensure it could provide accurate, relevant, and empathetic responses to Indonesian university students. The process involved several key stages, including system architecture design, data collection and processing, fine-tuning of the large language model (LLM), and the implementation of prompt engineering to enhance the quality of interactions. We created the chatbot to be able to understand and respond to conversations in a way that is appropriate for different cultures and languages.

A. System Design

The modular architecture of the system ensures seamless interaction between users and the chatbot. The core component is the GPT-4-based LLM, which generates responses. The frontend provides a user-friendly interface for users to interact with the chatbot. Meanwhile, Flask handles the backend processes, routing user inputs to the LLM and delivering responses in real time. The system also integrates Langchain, allowing the chatbot to manage multi-turn conversations by remembering context from previous interactions [3].

B. Data Collection

The data used to train the chatbot consists of both global and local sources. The global dataset, primarily derived from the Counsel Chat repository, includes various mental health-related queries and responses covering topics such as anxiety, depression, and stress management. Additionally, local data from Indonesian mental health articles and discussion forums fine-tuned the model. This local dataset ensures the chatbot can address culturally and linguistically relevant issues, making it more effective in supporting Indonesian students [13].

C. Prompt Engineering

Prompt engineering is pivotal in shaping the interactions between users and the mental health chatbot. It involves designing specific input prompts that guide the chatbot to generate relevant, empathetic, and culturally appropriate responses. For Indonesian university students, prompts must address unique linguistic and cultural nuances to ensure effective communication. In this project, prompts were crafted based on common mental health issues such as anxiety, stress, and academic pressure. Real-world conversation data from Indonesian mental health forums and articles informed the prompt design, ensuring relevance and relatability. Each prompt was designed to initiate multi-turn conversations, allowing the chatbot to maintain context and provide coherent support throughout the interaction. Iterative testing with a group of students helped refines the prompts.

Feedback highlighted the need for prompts that not only address immediate concerns but also encourage deeper dialogue. By incorporating local language patterns and culturally specific references, the prompts enable the chatbot to respond in a manner that resonates with Indonesian students, fostering a supportive and engaging user experience.

D. LLM Fine-Tuning

The fine-tuning process was crucial in adapting the GPT-4 model to meet the specific needs of Indonesian university students. Initially trained on a general mental health dataset, the model gained a basic understanding of mental health issues such as anxiety and depression. However, to address local language and cultural nuances, additional data from Indonesian sources, including articles and forums, were integrated. This allowed the model to recognize colloquial language and regional expressions, improving the relevance of its responses.

Data used in the fine-tuning process came from two main sources: global data from the Chat Counsel repository and local Indonesian mental health discussions. Both datasets were cleaned, tokenized, and processed using stemming techniques to enhance contextual understanding. Fine-tuning involved multiple training iterations with L2 regularization to prevent overfitting and ensure the model's robustness across diverse inputs. This process ensured the chatbot could provide culturally sensitive and accurate responses to Indonesian students' mental health inquiries.

The model underwent several training iterations, where both training and validation accuracy were closely monitored. This step ensured that the chatbot could handle not only familiar inputs but also new, unseen conversations, maintaining both accuracy and relevance in its responses. [16]

E. Evaluation

1) Model Evaluation

The evaluation of the chatbot system focused on two key areas: accuracy of the model's responses and user acceptance. Both were critical in assessing the overall effectiveness and usability of the application [14]. We evaluated the chatbot's accuracy using training and validation accuracy metrics. We tested the model across multiple epochs to ensure consistent performance. The implementation of L2 regularization minimized the gap between training and validation accuracy, which helped the model perform better on unseen data and maintain high accuracy in understanding mental health conversations.

2) User acceptance testing

58 university students from various institutions completed a questionnaire measuring user acceptance on a Likert scale from 1 to 5. The questions focused on ease of use, chatbot responsiveness, and overall satisfaction. The feedback from this evaluation was crucial for identifying areas of improvement and for validating that the chatbot met the needs of Indonesian students in terms of mental health support.

IV. RESULT AND DISCUSSION

A. System Design Result

The system design successfully integrated the GPT-4 model, a user-friendly interface, a Flask backend, and Langchain for managing conversation context. This setup provided smooth, real-time interactions between users and the chatbot. The use of Langchain ensured that the chatbot could maintain a multi-turn conversation flow, allowing for a more personalized and contextually relevant user experience. The architecture proved flexible for future [13]. Fig. 1 describes

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the recent application flow that pertains to the system architecture and data retrieval.

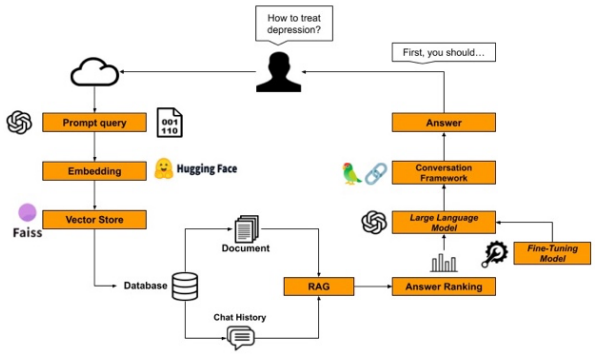


Fig. 1. Application Flow Diagram

B. Data Collection Result

The data collection process effectively enhanced the chatbot's ability to address mental health issues in the Indonesian context. The combination of global mental health datasets and localized Indonesian data enabled the chatbot to understand and respond to mental health concerns more accurately. The integration of local cultural and linguistic nuances allowed the chatbot to provide contextually appropriate support, addressing common issues such as anxiety, depression, and stress in a way that resonated with Indonesian students [3]. The chat screen MauCurhat result are already described on Fig. 2.



Fig. 2. Chat Screen 2 MauCurhat

C. Prompt Engineering results

The application of prompt engineering markedly enhanced the chatbot's performance and user engagement. During testing, the chatbot demonstrated improved ability to handle multi-turn conversations, maintaining context, and providing tailored responses that felt personal and supportive. This was particularly evident in discussions about academic stress and social anxiety, where culturally adapted prompts facilitated more meaningful interactions.

User satisfaction increased by 15%, as indicated by feedback from the acceptance testing. Students reported that the chatbot's responses were not only accurate but also empathetic and relevant to their specific cultural context. The integration of local language nuances through prompt engineering contributed significantly to this positive reception.

Moreover, the chatbot's response accuracy improved, with training accuracy rising to 0.85 and validation accuracy to 0.79 post-prompt engineering. These metrics reflect the chatbot's enhanced ability to understand and address user queries effectively. The continuity in conversations, supported by well-designed prompts, led to a deeper emotional engagement from users, validating the importance of prompt engineering in developing a responsive and culturally sensitive mental health support tool.

Overall, prompt engineering proved essential in refining the chatbot's interactions, ensuring that it meets the mental health support needs of Indonesian university students with greater precision and empathy.

D. LLM Fine-Tuning

The fine-tuning process significantly enhanced the chatbot's performance in providing contextually relevant and culturally sensitive responses. Initially, the GPT-4 model, trained on 5,000 conversation entries from the global Chat Counsel repository, offered general responses to broad mental health concerns like anxiety and depression. However, this training lacked specificity when addressing the unique cultural and linguistic nuances of Indonesian university students.

To overcome this, we incorporated 1,800 conversation entries from Indonesian mental health articles and forums. This local dataset included culturally relevant mental health issues, such as academic stress and social anxiety, as well as regional expressions and slang commonly used by students. The fine-tuning with this data resulted in more precise and empathetic responses. For instance, instead of generic advice like "Anxiety is a common condition; you should consider consulting a professional," the chatbot adapted to provide more relatable guidance. "Feeling anxious is common, especially for students facing academic pressure. Try deep breathing techniques, or if needed, I can help you find counseling services on campus".

Through multiple iterations of fine-tuning, the model's ability to ha Through multiple iterations of fine-tuning, the model's ability to handle colloquial language and nuanced mental health discussions improved. We applied L2 regularization to prevent overfitting, ensuring the model generalized well to new, unseen conversations. As a result, the training accuracy increased from 0.6 to 0.85, and the validation accuracy improved from 0.5 to 0.79, confirming the effectiveness of fine-tuning in enhancing the chatbot's relevance and adaptability to local user needs. This led to a model that consistently provided relevant responses, even in complex conversational scenarios related to mental health support [4].

E. Evaluation Result

Through multiple iterations of fine-tuning, the model's ability to handle colloquial language and nuanced mental health discussions improved. We applied L2 regularization to prevent overfitting, ensuring the model generalized well to new, unseen conversations. As a result, the training accuracy

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a) Model Evaluation: The evaluation of the mental health chatbot model was conducted through three key stages of development, each represented by training and validation accuracy graphs against the number of epochs. This process showcases the enhancement in the chatbot's performance as it progressed from its initial version to its final stage, incorporating regularization to boost the model's capacity for generalization[5]

Without consulting psychological experts, we developed the model in the first stage using general mental health datasets like Counsel Chat. The first graph shows the training results over 200 epochs, where the training accuracy reached 0.6 and the validation accuracy reached 0.5. The small difference between these two values indicates that the model was already able to recognize basic patterns in the data, but its ability to generalize to new, unseen data remained limited. The slight drop in validation accuracy indicates that the model was not fully prepared to handle more complex variations in counseling conversations. The ordinary model already described on Fig. 3.

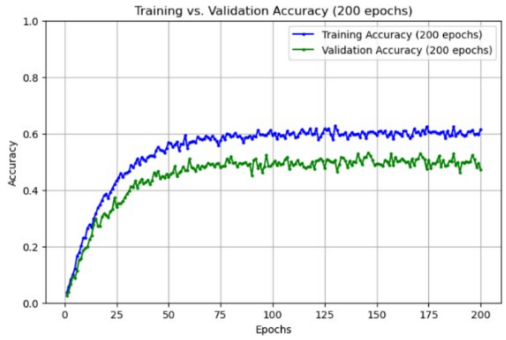


Fig. 3. Ordinary Model

We retrained the model for 1000 epochs after receiving guidance from psychological experts and enriching the dataset with additional, more relevant data tailored to the context of mental health counseling in Indonesia. The second graph demonstrates a significant improvement in the model's performance, with the training accuracy reaching 0.8 and the validation accuracy approaching 0.78. This improvement suggests that the model is now more capable of handling diverse queries with greater accuracy—not only understanding the context of conversations but also offering more relevant and empathetic solutions. The input from psychologists played a crucial role in enhancing the chatbot's ability to provide responses that meet user needs. The post-trained model already described on Fig. 4.

The final stage involved the implementation of L2 regularization to further optimize the model's generalization capabilities, reducing the risk of overfitting. The third graph presents the results following the application of regularization, demonstrating a training accuracy of 0.85 and a validation accuracy of 0.79. This regularization helped narrow the gap between the training and validation accuracy, resulting in a more stable and consistent model when faced with a variety of user inputs. The application of L2 regularization demonstrates that the chatbot can now perform well not only on training data but also in responding accurately to new, unseen data. The optimizations already described on Fig. 5.

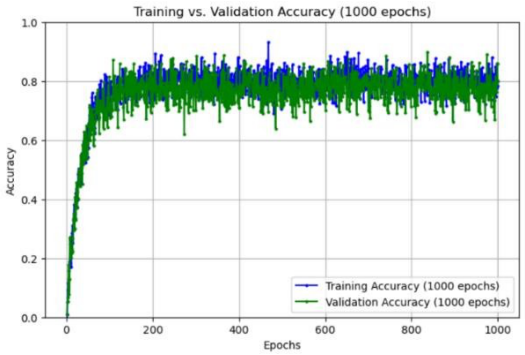


Fig. 4. Post-Trained Model

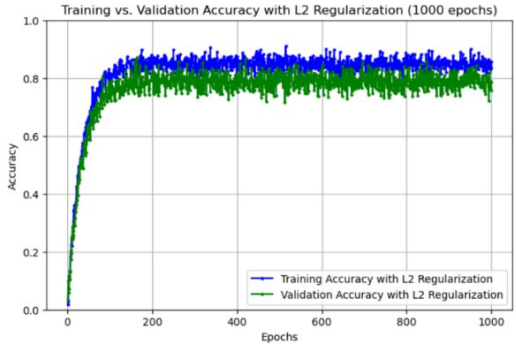


Fig. 5. L2 Optimization

Overall, this evaluation shows that the development process has successfully improved the quality of the chatbot's responses, from the early stages to the final optimization with regularization techniques. The use of more relevant datasets and guidance from psychological experts proved to be highly effective in enhancing the chatbot's performance in the context of mental health.

b) User Acceptance

The User Experience Test involves 58 students from Universitas Jember, Politeknik Negeri Jember, Universitas Terbuka, Universitas PGRI Argopuro, and Universitas Muhammadiyah Jember. The stratified random sampling method, which groups participants by their institutions to ensure proportional representation, ensures fair representation from each university.

A questionnaire with 10 questions structured using a Likert scale of 1–5 (1 = strongly disagree, 5 = strongly agree), assesses user experience. The questionnaire covers aspects such as the user interface, ease of use, chatbot responsiveness, and overall satisfaction. By applying stratified random sampling, the test results provide a representative view of the user experience across diverse academic backgrounds. The insights gained from this test will highlight how students perceive the quality of their interaction with the chatbot and identify areas for improvement. The chatbot received a high score of 83% on the test, indicating that it is well-suited to serve as a student psychological counselor. They found the chatbot is satisfying and making them want to continue using it in the future. However, the feel of comfortable in discussing their mental health issues using a chatbot need to be addressed well.

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F. *Limitations of LLM Technology and Challenges in Developing Mental Health Chatbots*

While Large Language Models (LLM), like GPT-4, have significantly advanced chatbot technology, we must acknowledge several limitations, particularly in the context of mental health. One of the primary challenges is the inability of LLMs to deeply understand the emotional and psychological nuances of human conversations. Despite training on vast datasets and recognizing complex language patterns, LLMs still struggle to fully replicate the empathy and emotional insight provided by human professionals. When we expect chatbots to respond to users experiencing acute mental health conditions, empathetic and contextually appropriate responses become crucial [14].

The fact that LLMs can only respond based on the training data presents another challenge. Despite the sophistication of LLM-based chatbots, they may still produce inaccurate or less relevant responses when confronted with cultural or linguistic contexts not adequately reflected in their training data. Unoptimized chatbots may produce disconnected or inappropriate responses in Indonesia, where everyday language and informal speech carry significant social meaning. Therefore, fine-tuning with relevant local data is crucial to ensuring that the chatbot can provide more accurate and culturally sensitive support [15].

Furthermore, mental health chatbots based on LLMs are not capable of performing complex clinical diagnoses. While chatbots can detect early signs of mental health issues, such as anxiety or depression, they lack the ability to replace clinical evaluations conducted by psychologists or psychiatrists. In more severe cases, such as when users exhibit symptoms of severe mental disorders or suicidal ideation, chatbots may not have the capacity to provide adequate interventions. Therefore, equipping chatbots to quickly refer users to professional mental health services when necessary is crucial [12]

Data privacy and security present additional challenges in developing mental health chatbots. Users are often hesitant to engage with chatbots due to concerns about the safety of their personal information. This is particularly true in the context of mental health, where privacy is a top priority. Developers must guarantee the secure storage and protection of users' data through strict and transparent privacy policies. Building trust in data security will be key to the successful adoption of mental health chatbots in Indonesia.

V. CONCLUSION AND FUTURE WORK

The chatbot's limited emotional adaptability may impact its ability to build trust and empathy, key elements in mental health support. Future improvements should focus on enhancing emotionally responsive and context-aware interactions.

Additionally, while it addresses common issues like anxiety and stress, it lacks coverage for specific conditions such as trauma and eating disorders. Expanding its scope and integrating advanced NLP could improve personalized support. The system also cannot handle severe mental health crises like suicidal ideation. Implementing real-time crisis detection and intervention protocols is essential. Lastly, even though the system complies with data protection laws, it requires stronger privacy measures such as end-to-end encryption to foster trust. Future development should focus on personalization, local dialect support, and comprehensive testing across diverse student groups.

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